

OverallNet: Scale-Arbitrary Lightweight SR Model for handling 360° Panoramic Images

DongSik Yoon
R&D Center AI LAB, HDC LABS
South Korea
kevinds1106@hdc-labs.com

Jongeun Kim
R&D Center AI LAB, HDC LABS
South Korea
JongeunKim@hdc-labs.com

Seonggeun Song
R&D Center AI LAB, HDC LABS
South Korea
ssg97@hdc-labs.com

Yejin Lee
R&D Center AI LAB, HDC LABS
South Korea
yejin.lee@hdc-labs.com

Gunhee Lee
R&D Center AI LAB, HDC LABS
South Korea
Gunhee_Lee@hdc-labs.com

Abstract

Estimating room layouts from 360° panoramic images is an essential task in computer vision, enhancing 3D scene understanding from 2D images. To achieve and handle high-quality panoramas necessitates a Super-Resolution (SR) technique that is both light and capable of arbitrary-scale SR, optimized for efficient inference. In this study, we propose a simple model that employs a modular arbitrary-scale technique. Additionally, our model incorporates quantization to maximize efficiency during user inference, making it well-suited for processing high-quality panoramic images.

Keywords

Super Resolution, Arbitrary Scaling, 2D Quantization, Panoramics

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1 Introduction

The task of interpreting 3D scenes from 2D images is crucial in computer vision, as it plays a vital role in research areas such as augmented and mixed reality. One specific task, reconstructing floor plans from panoramas, relies on a relatively small set of real images, leading to potential losses during the remapping process. To address these challenges, converting images taken with 360° cameras to high resolution is essential. This process particularly requires a Super-Resolution (SR) technique that is lightweight, capable of arbitrary-scale SR, and optimized for inference.

OverNet [Behjati et al. 2021] specifically handles the need for multi-scale SR by utilizing a lightweight overscaling network. The model proposes the Overscaling Module (OSM) which reduces artifacts and enhances SR clarity by downscaling multiple overscaled

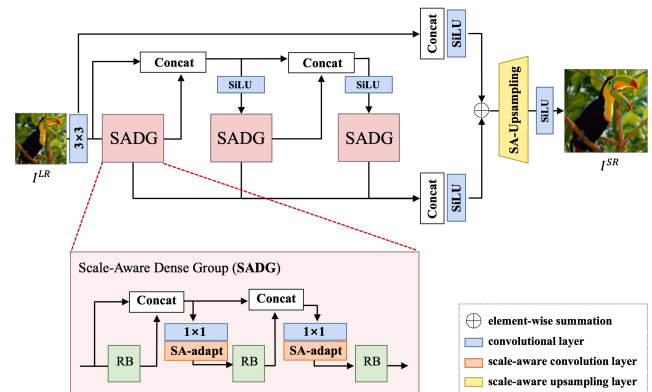


Figure 1: Illustration of the proposed framework and scale-aware Dense Group (SADG) block.

values generated for the same pixel. However, OverNet excels only at the scales corresponding to the training dataset, as its structural limitations within OSM lead to reduced SR performance at different scales. Therefore, our objective is to alleviate the limitations of the current OverNet and propose a lightweight, scale-independent SR model named OverallNet using various techniques. To this end, first, we adopt the scale-aware module proposed in ArbSR [Wang et al. 2021] and introduce a new framework that includes the SADG block. Then, we integrate gradient variance loss [Abrahamyan et al. 2022] as our overall loss function, enhancing the edges of generated images with a gradient map. Finally, we apply 2DQuant [Liu et al. 2024] to reduce memory usage and ensure effectiveness in real-world applications.

2 Method

In the case of OverNet [Behjati et al. 2021], the structure of the Overscaling Module (OSM) constrains overscaling to a factor larger than the target, followed by bicubic downscaling to attain super-resolution at the desired scale. This methodology presents a significant drawback: the image sharpness diminishes when the scale factor of the target data is not consistent with that of the training data. For example, when OverNet attempts super-resolution at a scale of $\times 2$, using a model trained at a scale of $\times 4$, the image is excessively overscaled to a factor like $\times 5$. It is then downscaled and combined. This process results in images with blurred or less defined edges, leading to a loss of significant details.

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Table 1: Quantitative comparison of PSNR metrics on Set5 and BSD100 datasets.

Method	Set5			BSD100		
	$\times 2$	$\times 3$	$\times 4$	$\times 2$	$\times 3$	$\times 4$
Bicubic	33.66	30.39	28.42	29.56	27.21	25.96
OverNet $\times 2$	38.11	32.05	30.16	32.22	28.11	26.78
OverNet $\times 3$	37.44	34.46	31.97	1.95	29.11	27.50
OverNet $\times 4$	36.24	33.89	32.25	31.34	28.90	27.59
Ours	37.87	34.38	32.22	31.95	29.00	27.57
Ours (8bit)	37.55	34.14	32.02	31.76	28.86	27.46

2.1 Proposed Architecture

To address the aforementioned issues of OverNet, we propose a novel architecture that incorporates the scale-aware module of ArbSR [Wang et al. 2021], as shown in Figure 1. ArbSR is a plug-in module that enables existing SR models to perform scale-arbitrary super-resolution by using conditional convolutions to dynamically generate filters based on input scale information, allowing the network to adapt to specific scale factors. To further enhance super-resolution sharpness and realism, we introduce a feature extractor built with a block called the scale-aware dense group (SADG). Unlike the existing OSM, our SADG block includes a scale-aware upsampling layer, facilitating scale-arbitrary SR at various factors. Moreover, based on recent research that the Swish activation function outperforms the ReLU activation function in SR tasks, we replace all activation functions in the SADG block with SiLU layers. Additionally, to better preserve detailed edges, we adopt a gradient variance loss function [Abrahamyan et al. 2022] along with the previously used L1-based multi-scale loss. This combination ensures that super-resolution images retain high-quality edge details and overall visual fidelity.

2.2 2D-Quantization

For 2DQuant [Liu et al. 2024], a two-stage post-training quantization technique, tailored for the SR domain, is employed to optimize the clipping range for weights and activations. Initially, weights and activations are quantized by dividing them into uniform intervals while gradually narrowing the clipping range. Subsequently, bounds that yield the lowest L2 distance between the quantized model and the full-precision (FP) model are explored.

Afterward, we designate the FP model as the teacher and the quantized model as the student, aiming to minimize the output discrepancies of each layer to determine a more precise clipping range. Opposed to the 2DQuant, we improve the training process's stability by employing distillation at the SADG block level rather than at the layer level. This method effectively reduces the disparity in the generated feature maps.

3 Experiments

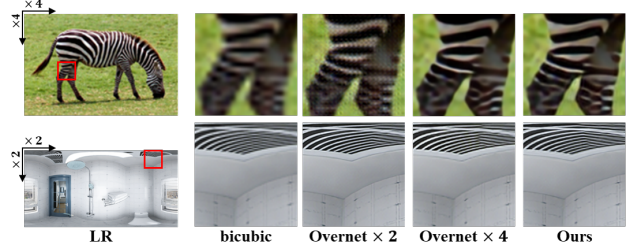
To demonstrate the effectiveness of our proposed method, we compare it with the existing OverNet, which is pretrained with various scale factors. For this evaluation, we use a well-established metric, Peak Signal-to-Noise Ratio (PSNR), and conduct tests on the Set5 and BSD100 datasets.

As illustrated in Table 1, the proposed method maintains robust performance across all scales using a single model. In contrast, OverNet exhibits significant degradation in results when applied to

Table 2: Quantitative comparison of PSNR metrics on Structured3D and 360-SOD.

Method	Structured3D					
	$\times 1.2$	$\times 1.6$	$\times 2.4$	$\times 2.8$	$\times 3.2$	$\times 3.6$
OverNet	33.66	33.36	31.15	30.18	29.37	28.71
Ours	35.18	32.92	31.15	30.62	30.05	29.56

Method	360-SOD					
	$\times 1.2$	$\times 1.6$	$\times 2.4$	$\times 2.8$	$\times 3.2$	$\times 3.6$
OverNet	34.64	33.95	31.67	30.77	30.02	29.39
Ours	35.77	33.77	31.65	30.94	30.34	29.80

**Figure 2: Qualitative comparison of the existing OverNet with ours.**

scales not included in the training dataset. For example, our single model surpasses OverNet $\times 2$ method results on Set5 dataset at both $\times 3$ and $\times 4$ scale. In addition, the quantized version of our model achieves optimization with minimal performance loss, underscoring its efficiency and practicality for real-world applications.

To further validate the effectiveness of our method in handling panoramic images, we conduct additional quantitative comparisons using the Structured3D and 360-SOD datasets, randomly selecting 200 panoramic images from each dataset, respectively. As indicated in Table 2, our proposed method outperforms the existing OverNet and successfully performs SR at non-integer scales. Especially, our proposed method also outscores 4 out of 6 scales in both datasets.

4 Future Work

In this paper, we adopt various techniques which improve the performance of SR models such as enhancing the loss function, enabling SR at arbitrary scale factors, and quantization to utilize the model effectively. These strategies could be integrated as a whole and become a model-independent technique. In future work, we are interested in leveraging these techniques to upgrade existing SR models in directions that users desire.

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